

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 5: System Software

Subtopic 5.2: Language Translators

Concept 5.2.3: Integrated Development Environment (IDE) Features

Total Questions: 11

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_11 (Q5.c)

- 5 The Operating System (OS) is responsible for hardware and security management in a computer system.

- (c) Identify **two** presentation features found in a typical Integrated Development Environment (IDE).

1

2 [2]

Question 2 | Source: 9618_w24_qp_11 (Q4.d.i)

- 4 A student uses a laptop to write a program that is saved as a text file.

- (d) The program code is written using an Integrated Development Environment (IDE).

- (i) One presentation feature found in a typical IDE is prettyprint.

Identify **and** describe **one other** presentation feature found in a typical IDE.

Feature

Description

.....

..... [2]

Question 3 | Source: 9618_w24_qp_11 (Q4.d.ii)

- 4 A student uses a laptop to write a program that is saved as a text file.

- (d) The program code is written using an Integrated Development Environment (IDE).

- (ii) One debugging feature found in a typical IDE is single stepping.

Identify **and** describe **one other** debugging feature found in a typical IDE.

Feature

Description

.....

..... [2]

Question 4 | Source: 9618_w24_qp_12 (Q4.a)

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

(a) Describe how the programmers can use the debugging features of a typical IDE during the development of the program.

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..... [4]

Question 5 | Source: 9618_s24_qp_12 (Q8.b)

8 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

- (b) IDEs have many features other than built-in translators.

Complete the table by identifying **one other** common IDE feature that can be used for each purpose. Describe how each feature helps the user during program development.

Each feature must be different. Do **not** give translator as one of your features.

Purpose	IDE feature	Description
for coding		
for presentation		
for debugging		

[6]

Question 6 | Source: 9618_w23_qp_11 (Q6.c.i)

- 6 A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

- (c) (i) Identify **two** features that support the visual presentation of the code in a typical Integrated Development Environment (IDE).

- 1
.....
- 2
.....

[2]

Question 7 | Source: 9618_w23_qp_11 (Q6.c.ii)

6 A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

(ii) Identify **two** features that support the debugging of the code in a typical IDE.

- 1
-
- 2
-

[2]

Question 8 | Source: 9618_w23_qp_13 (Q5.a)

5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

(a) Describe the following features of a typical IDE.

- Context-sensitive prompts
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-
-
- Single stepping
-
-
-
-

[4]

Question 9 | Source: 9618_s23_qp_12 (Q7.c)

7 A software developer is working in a team writing a program for a client.

- (c) The development team uses an Integrated Development Environment (IDE).

Complete the table by describing the typical features found in an IDE.

Feature	Description
Breakpoints	
Dynamic syntax checks	
Context-sensitive prompts	
Single stepping	

[4]

Question 10 | Source: 9618_w21_qp_11 (Q4.b.ii)

- 4 Francis is starting his first job as a software developer for a multinational company.

- (b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

- (ii) A typical IDE provides debugging tools to support the testing of a program.

Identify **three** other tools or features found in a typical IDE to support the writing of the program.

1

2

3

[3]

Question 11 | Source: 9618_s21_qp_12 (Q7.b.ii)

7 Jennifer is writing a computer program for her A Level homework.

(b) Jennifer uses an Integrated Development Environment (IDE) to write her computer program.

(ii) Identify **two** debugging tools that a typical IDE can provide.

- 1
-
- 2
-

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q5.c)

5(c)	1 mark per bullet point, max 2 marks • Prettyprint • Expanding and collapsing code blocks • <u>Automatic</u> / <u>smart</u> indentation / formatting	2
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Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q4.d.i)

4(d)(i)	1 mark for correct name, 1 mark for matching description • Expand/collapse code blocks... • ... sections of source code that are part of the same block can be expanded to see the content or collapsed so that the overall code is seen • Auto-indentation // auto-formatting... • ... automatically indents/formats code as the user types so that the structure is clear // aids readability	2
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Mark Scheme for Question 3 | Source: 9618_w24_ms_11 (Q4.d.ii)

4(d)(ii)	1 mark for correct name, 1 mark for matching description • Breakpoints ... • ... stops the code running on a set line to view the current status/variable contents/program flow • Report window // variable watch window ... • ... shows the values in variables/data structures and how they change when each line is run	2
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Mark Scheme for Question 4 | Source: 9618_w24_ms_12 (Q4.a)

4(a)	1 mark for each bullet point (max 4) e.g. • Single stepping • Run the program one line at a time • ... and check the variable contents / program flow // show the effect of each line of code • Set breakpoints • ... run the code up to a set line • ... and then check the status • Variable/report watch window • ... view how the data changes as the program is running	4
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Mark Scheme for Question 5 | Source: 9618_s24_ms_12 (Q8.b)

8(b)

1 mark for identification of each feature and 1 mark for matching description:

6

e.g.

For coding:

IDE feature	Description
Context-sensitive prompts	Gives suggestions for code as the user types instead of having to write/remember the code
Auto-correct	Corrects spelling mistakes so that user has fewer errors to correct

For presentation:

IDE feature	Description
Pretty-printing	Colour code keywords so the user can identify any errors
Expand/collapse (code) blocks	The user can hide code that they are not currently working on

For debugging:

IDE feature	Description
Single stepping	Run the code one line at a time // shows the effect of each line of code
Breakpoints	Stop the code running at a set point to check the flow/variable contents

Mark Scheme for Question 6 | Source: 9618_w23_ms_11 (Q6.c.i)

6(c)(i)	1 mark for each bullet point (max 2) • Prettyprint • Expand/collapse code blocks • Auto indentation / formatting	2
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Mark Scheme for Question 7 | Source: 9618_w23_ms_11 (Q6.c.ii)

6(c)(ii)	1 mark for each bullet point (max 2) • Single stepping • Breakpoints • Report window • Variable expressions	2
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Mark Scheme for Question 8 | Source: 9618_w23_ms_13 (Q5.a)

5(a)	1 mark for each bullet point (max 2 for each feature). Context-sensitive prompts: • As the code is being written • ...the options to complete the statement are shown Single stepping: • allows the programmer to execute the program one line at a time • ...so that the effects of each statement can be seen	4
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Mark Scheme for Question 9 | Source: 9618_s23_ms_12 (Q7.c)

7(c)	1 mark each Breakpoints: • Stop the code at a specific line to check the current progress / values Dynamic syntax checks: • Highlight / underline / colour syntax errors as the code is entered Context-sensitive prompts: • Suggest the code to add // automatically complete statements Single stepping: • Run the code one line at a time so the values can be checked	4
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Mark Scheme for Question 10 | Source: 9618_w21_ms_11 (Q4.b.ii)

4(b)(ii)	1 mark for each correct tool e.g. • Colour coding // pretty printing • Auto-complete • Auto-correct • Context sensitive prompts • Expand and collapse code blocks	3
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Mark Scheme for Question 11 | Source: 9618_s21_ms_12 (Q7.b.ii)

7(b)(ii)	1 mark per correct tool to max 2 e.g. • Breakpoints • Single stepping • Report windows	2
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